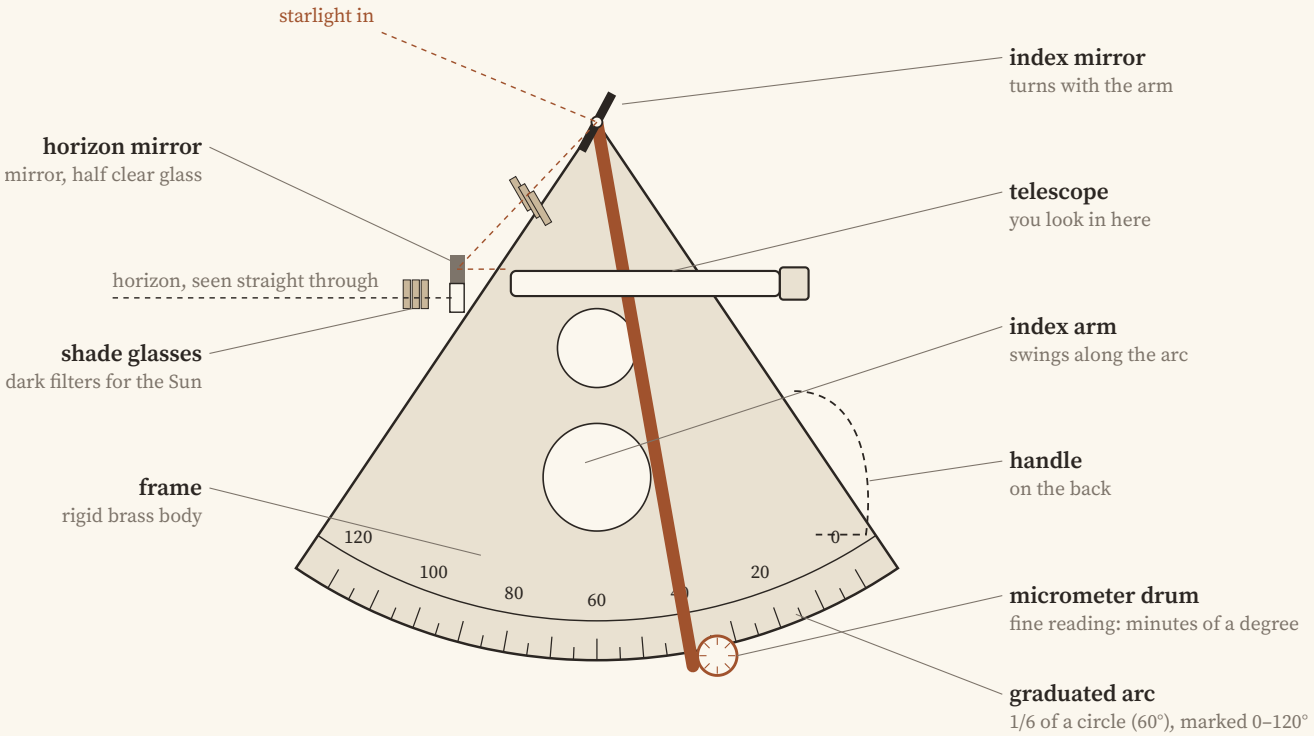
The sextant: how old sailors found their latitude

A hand-held tool for measuring one angle: how high a star or the Sun stands above the sea horizon.

1 What a sextant looks like



Why “sextant”? The arc is one-sixth of a circle (60°). Yet it reads up to about 120°, because the two mirrors double the angle the arm turns.

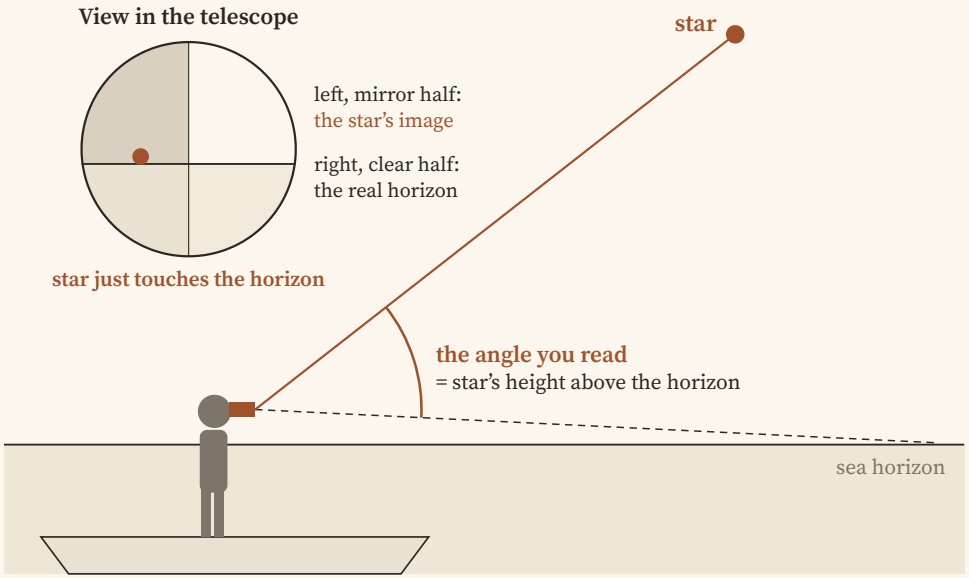
The one moving part is the **index arm**. It turns the index mirror on its pivot at the top.

The arc gives whole degrees. **The drum** adds the fine part, in minutes (1 minute = 1/60 of a degree).

Arm shown set at 40°.

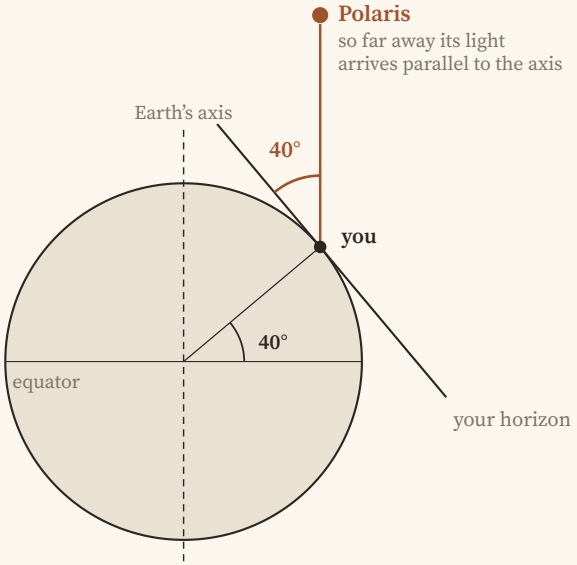
2 Taking a sight

Look at the horizon through the clear half of the horizon mirror. Swing the index arm until the star's image, seen in the mirror half, just touches the horizon. Read the angle off the arc.



3 Polaris's height ≈ your latitude

Latitude: how many degrees you are north (or south) of the equator.



Small correction. Polaris sits about 0.6° from the exact pole today, so navigators added a small correction from a printed table.

By day: the Sun at noon. Measure the Sun at its highest point; latitude = 90° − that height, adjusted by the Sun's seasonal position from a printed almanac.

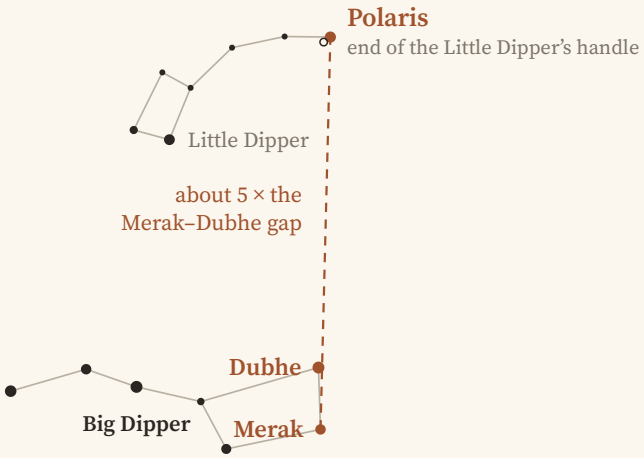
Latitude was routine. Longitude (how far east or west) needed an accurate clock, which came later.

Finding the sky's pole to measure

The whole sky turns around one point above each of Earth's poles. Its height above the horizon is your latitude. Star patterns lead you to it. Patterns shown at true relative positions; through the night they turn around the pole.

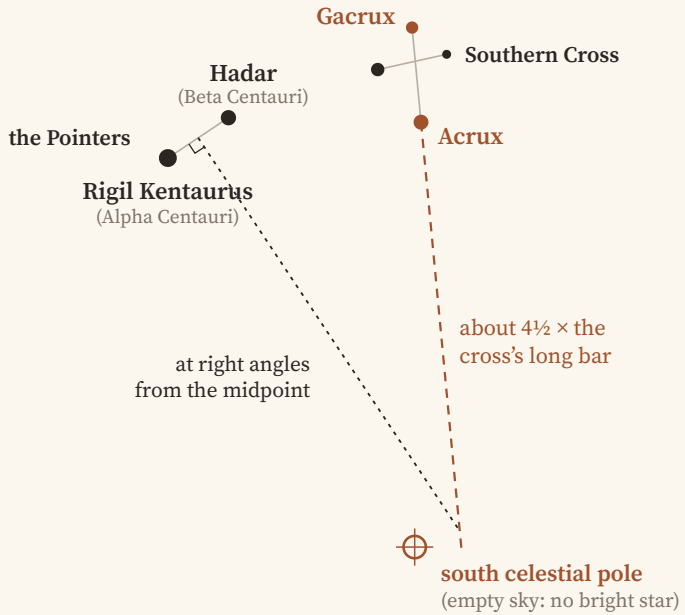
4 North: follow the pointers

Go from Merak through Dubhe, the two stars at the front of the Big Dipper's bowl, and keep going about 5 times their spacing. You reach Polaris, alone at the end of the Little Dipper's handle.



5 South: no pole star, so find the spot

Extend the Southern Cross's long bar, Gacrux to Acrux, about 4½ times its length. Check it: from the midpoint of the Pointers, go off at right angles. Where the two lines meet is close to the pole.



Accuracy. Both southern tricks land about 3° from the true pole, and the sailor must measure the height of an empty spot of sky, so a southern fix was less precise than one from Polaris.

Today. GPS does the same job with timed signals from satellites.

Sources. Star positions J2000 (Hipparcos/HYG). Methods: *The American Practical Navigator* ch. 16; ASSA "Celestial direction finding"; Te Ara (NZ). Polaris offset 0.62° (2026).